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ASSESSMENT TOOL - 1

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QUESTION - 1

TCP THREE - WAY HANDSHAKE

TCP uses a three - way handshake to establish a reliable connection between a client and a server.

1) SYN: The client sends a SYN packet to the server to request a connection and provides its initial sequence number

2) SYN - ACK - The server responds with SYN and ACK. It acknowledge the clients sequence number and sends its own sequence number.

3) ACK : The client sends an ACK to acknowledge the server's sequence number.

After these three steps, the TCP connection is established and data can be transformed .

QUESTION - 2

DNS USING UDP

• DNS commonly uses UDP because DNS requests are usually small and need to be processed quickly

• The UDP header is only in 8 bytes, whereas TCP has a minimum header size of 20 bytes

• UDP doesnt contain sequence numbers acknowledgement numbers or connection setup. This reduces overhead and makes DNS communication faster and however, UDP doesnt garranty delivery.

→ If a DNS packet is lost. the application can retry the request.

→ Therefore UDP provides higher speed and lower overhead, while TCP provides greater reliability.

QUESTION - 3

TCP VS UDP DURING PACKET LOSS

⇝ When packet loss occurs, TCP and UDP behaves differently

⇝ TCP detects missing packet using acknowledgements and sequence numbers.

⇝ If a packet is lost, TCP retransmits it

⇝ This ensures that all data reaches destination correctly and in proper order.

⇝ UDP doesnt retransmit lost packets. It has no built in acknowledgement or retransmission mechanism.

⇝ UDP is faster but less reliable, which is useful for applications such as live streaming and voice / video.

TCP Vs UDP During Packet loss.

PACKET LOSS

TCP

DETECTS LOST PACKET

RETRANSMITS PACKET

DATA DELIVERED RELIABLY

MORE RELIABLE

UDP

NO LOSS DETECTION

PACKET IS DISCARDED

DATA MAY BE INCOMPLETE

FASTER

QUESTION - 4

DNS QUERY TYPES AND RESOLUTION

- A record : Provides the IPV4 address of a domain.
- AAA record : Provides the IPV6 address.
- MX record : Provides information about the mail server

USER / CLIENT

$\downarrow$ DNS QUERY

LOCAL DNS RESOLVER

$\downarrow$ IF NOT CATCHED

ROOT DNS SERVER

$\downarrow$

TLD DNS SERVER (.com, .org)

$\downarrow$

AUTHORISATION

$\downarrow$ IP ADDRESS

LOCAL DNS RESOLVER $\xrightarrow{\text{DNS RESPONSE}}$ CLIENT